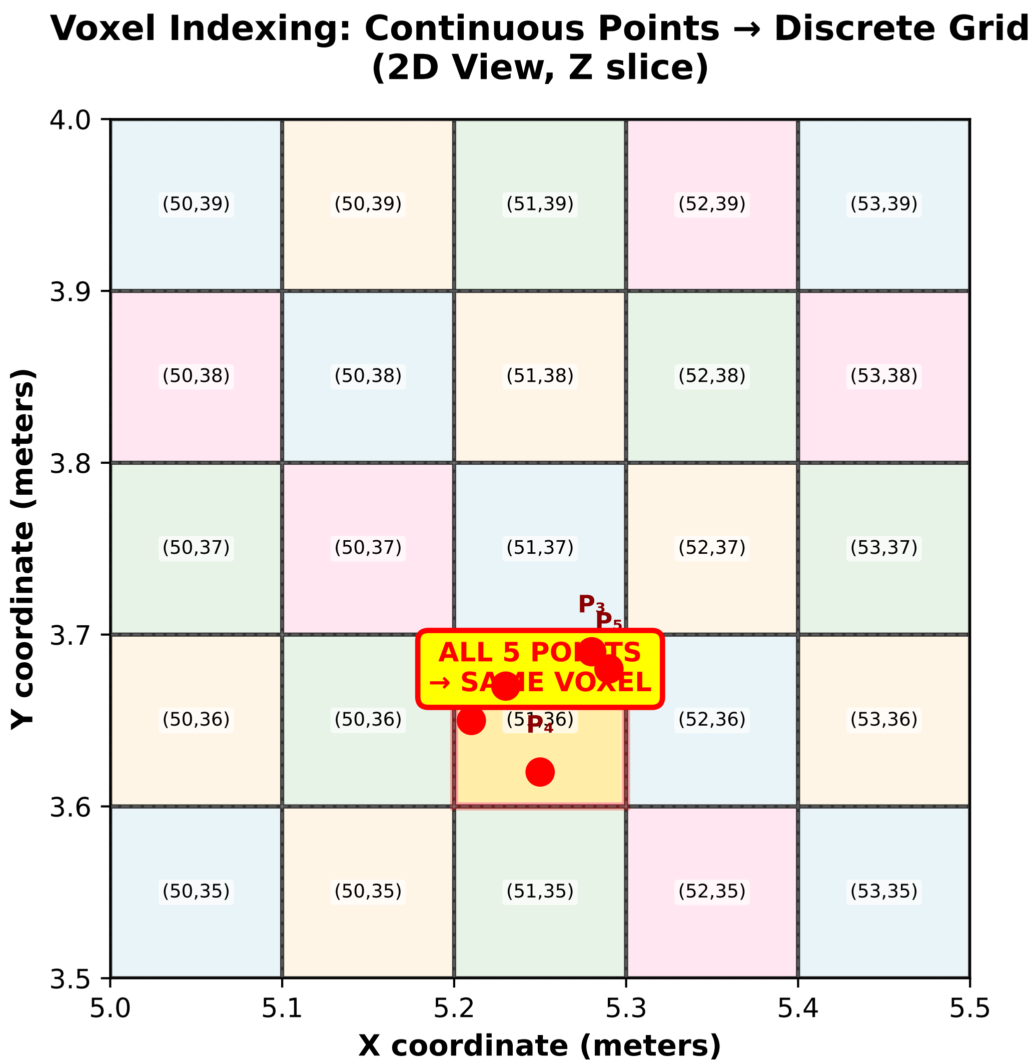
Voxel Indexing: How Floor Division Maps Continuous Space to Discrete Grid



Voxel Index Computation

$$\textbf{Voxel Index} = \left\lfloor \frac{\textbf{Point Position}}{\textbf{Voxel Size}} \right\rfloor$$

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Point P₁:            (5.23, 3.67, 1.42)
Voxel size:          σ = 0.10 m

Calculation:
  i_x = ⌊5.23/0.10⌋  = ⌊52.3⌋ = 52
  i_y = ⌊3.67/0.10⌋  = ⌊36.7⌋ = 36
  i_z = ⌊1.42/0.10⌋  = ⌊14.2⌋ = 14

Result:              Voxel Index = (52, 36, 14)
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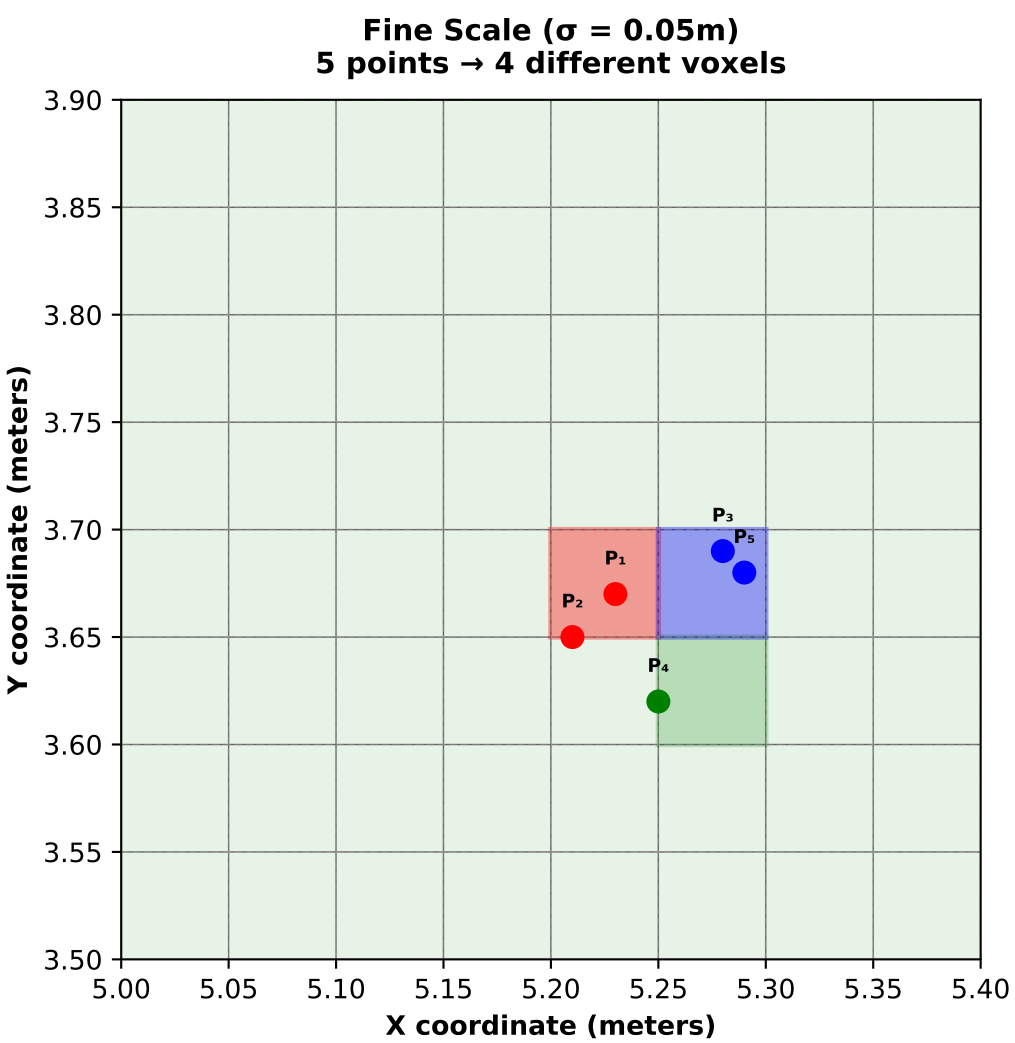

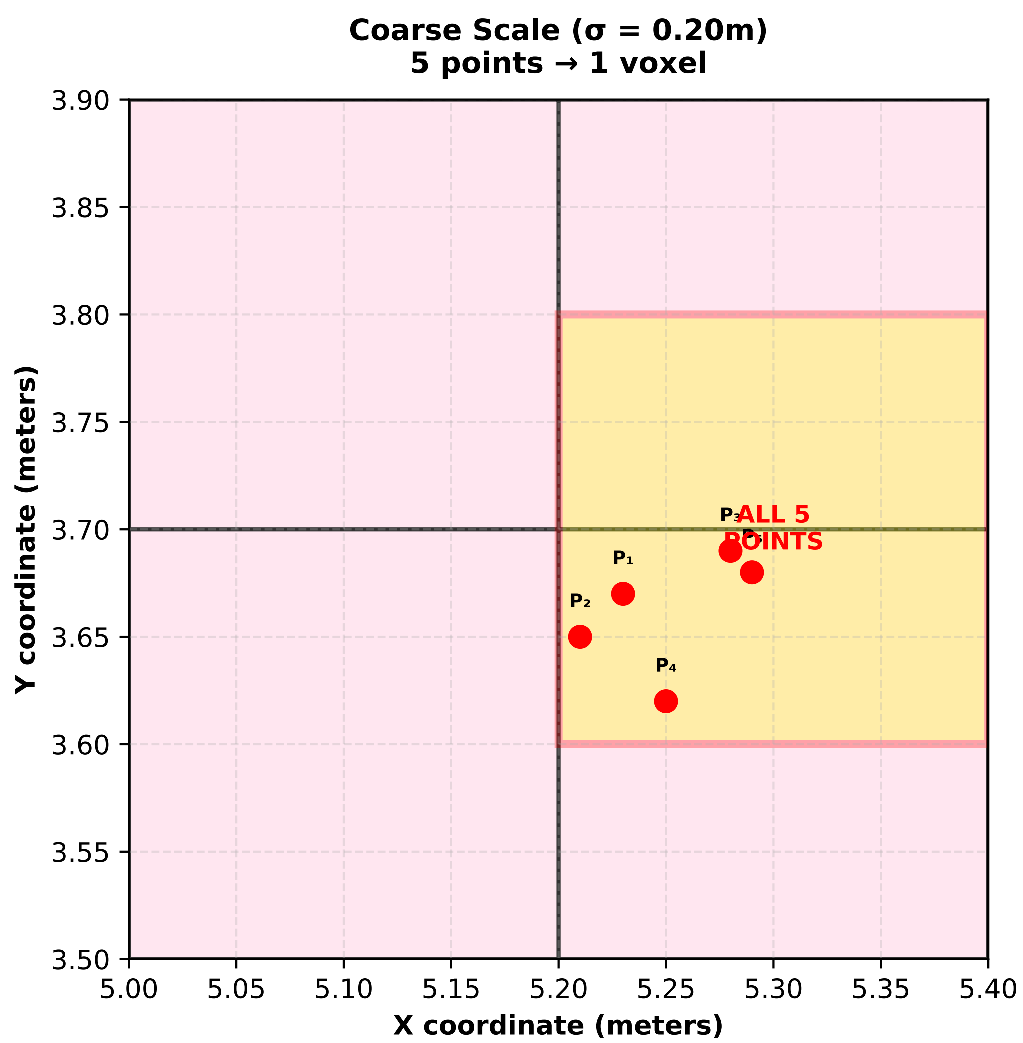

How Voxelization Generalizes

Fine Scale (σ=0.05m)
- 5 points → 4 voxels
- More detail
- More sparse
- Better boundaries

Medium Scale (σ=0.10m)
- 5 points → 1 voxel
- Balanced
- Average density
- Standard choice

Coarse Scale (σ=0.20m)
- 5 points → 1 voxel
- Less detail
- More dense
- Efficient

Key Insight: Voxel size controls the level of generalization

VoxAdapt learns optimal scale per-point instead of using fixed scale!